

Interpretable Diabetes Risk Analysis Using Bayesian Networks

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Abstract

This mini-project investigates the use of Bayesian Networks to model and explain diabetes risk using clinical variables from the Pima Indians Diabetes dataset.

We compare different preprocessing strategies and Bayesian Network structures, including manually designed models and structures learned from data, and perform probabilistic inference to analyze the effect of evidence on diabetes risk.

We found that preprocessing choices strongly influence model behavior and interpretability, with learned structures combined with appropriate discretization producing more stable and meaningful posterior inferences.

Introduction

Domain

This project focuses on modeling diabetes risk using a Bayesian Network built from clinical and demographic variables. Diabetes is a chronic condition influenced by multiple interacting factors, making it a suitable domain for probabilistic graphical models that explicitly represent dependencies among variables and allow interpretable inference.

The analysis is based on the Pima Indians Diabetes dataset (Smith et al. 1988), which contains medical measurements such as glucose concentration, body mass index, age, and pregnancy count, along with a binary outcome indicating diabetes diagnosis. The dataset has been widely used in the literature for studying diabetes prediction and presents typical challenges such as missing or implausible values and the need for discretization of continuous variables.

Rather than aiming for maximum predictive accuracy, this work emphasizes interpretability and probabilistic reasoning. Bayesian Networks are employed to examine how evidence on specific variables influences the posterior probability of diabetes and to assess how modeling and preprocessing choices affect inference behavior.

Aim

The aim of this project is to explore the use of Bayesian Networks for interpretable diabetes risk analysis.

In particular, we investigate how different preprocessing

choices and network structures influence probabilistic inference results, focusing on the stability and plausibility of posterior diabetes risk estimates rather than predictive accuracy.

Method

The project was implemented using the `pgmpy` (Ankan and Panda 2015) library to construct Bayesian Network models and perform probabilistic inference. The analysis started with data preprocessing, including handling missing or implausible values (e.g., using `scikit-learn` imputation methods (Pedregosa and others 2011)) and discretizing continuous variables using different strategies.

Several Bayesian Network structures were considered, including manually specified models and structures learned from data using tree-based search and score-based hill-climbing algorithms with different scoring functions. For each model, conditional probability distributions were estimated from the processed dataset.

Inference queries were then performed to evaluate how evidence on selected variables affects the posterior probability of diabetes. By varying preprocessing methods and network structures, we analyzed their impact on inference behavior and model interpretability.

Results

The experiments show that Bayesian Network behavior is strongly affected by preprocessing and structure choices. Simple or manually defined structures often produced unstable or uninformative posterior probabilities, whereas models learned from data yielded more coherent inference results.

In particular, combinations of appropriate imputation and discretization methods led to posterior diabetes risk estimates that were more consistent and interpretable across different evidence configurations.

Model

Figure 1 illustrates a representative Bayesian Network used in this project. The model includes nodes corresponding to clinical and demographic variables such as glucose concentration, body mass index, age, and number of pregnancies, together with a binary node representing diabetes diagnosis.

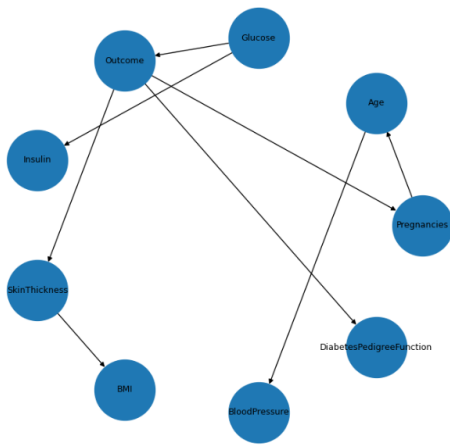


Figure 1: Bayesian Network structure learned using hill-climbing structure learning with AIC score, after KNN imputation and KMeans discretization.

Continuous variables were discretized into a limited number of states to enable parameter learning and tractable inference. Conditional probability tables were estimated directly from the preprocessed dataset, following imputation of missing or implausible values.

Different modeling choices were explored. In addition to manually specified dependency assumptions and tree-based searches, score-based structure learning was applied using hill-climbing algorithms with alternative scoring functions. This allowed us to analyze how structural assumptions and preprocessing decisions influence the inferred conditional dependencies and posterior diabetes risk estimates. Further implementation details are provided in the accompanying notebook.

Analysis

Experimental setup

The analysis is based on probabilistic inference queries aimed at evaluating how evidence on selected clinical variables affects the posterior probability of diabetes. For each Bayesian Network model, we queried the probability of the diabetes node under different evidence configurations, such as high or low glucose levels and varying body mass index ranges.

Experiments were repeated across different preprocessing pipelines and network structures to assess the robustness of inference results. The evaluation focused on qualitative comparison of posterior probability trends and their plausibility rather than on predictive accuracy metrics. In particular, we examined whether changes in evidence produced consistent and interpretable variations in diabetes risk estimates.

Results

The inference results highlight substantial differences between models depending on preprocessing and structural assumptions. Bayesian Networks with simple or manually

defined structures often produced posterior diabetes probabilities that were either insensitive to evidence or highly unstable across similar queries.

In contrast, models learned from data exhibited more coherent behavior. In these cases, increases in glucose levels and body mass index generally led to higher posterior diabetes risk, aligning with domain expectations. However, some sensitivity to discretization boundaries remained, indicating that inference outcomes are influenced not only by network structure but also by preprocessing choices.

Overall, the results confirm that careful preprocessing and structure learning are essential for obtaining meaningful and interpretable probabilistic inferences in this domain.

Conclusion

This project explored the use of Bayesian Networks for interpretable diabetes risk analysis, with particular emphasis on the impact of preprocessing and structural choices on probabilistic inference. The results show that inference quality and stability depend strongly on how variables are discretized and how the network structure is defined or learned. While the models are not intended to provide clinical predictions, they demonstrate how Bayesian Networks can support transparent reasoning about medical risk factors. Limitations include sensitivity to preprocessing decisions and the use of a single dataset, suggesting that future work could explore alternative discretization strategies, additional variables, or external data sources.

Links to external resources

GitHub repository: github.com/sara-tajernia/Diabetes-Analysis-with-Bayesian-Network

References

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